

Experiment #02

Linux Shell commands for performing directory/files operations

Objective

The objectives of this lab are to familiarize students how to perform:

1. File and directory creation
2. File and Directory deletion
3. File and Directory moving from one location to another
4. File and Directory copying from one location to another
5. Displaying data of file

Software Tools

- Ubuntu 16.04
- GCC Compiler
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Theory

Why Linux:

- ✓ LINUX is free.
- ✓ Can view and edit the source code of OS
- ✓ It is fully customizable.
- ✓ Most Important Feature is Stability
- ✓ Important in shared environments and critical applications
- ✓ LINUX has better security structure.
- ✓ High Portability
- ✓ Easy to port new H/W Platform
- ✓ Written in C which is highly portable

Layers in Linux:

Three important parts of Linux are Kernel, Shell and file system.

Kernel:

- ✓ Kernel is the heart of the operating system.
- ✓ It is the low level core of the System that is the interface between applications and H/W.
- ✓ Functions are: Manage Memory, I/O devices, allocates the time between user and Process, inter process communication, sets process priority.

Shell:

- ✓ The shell is a program that act as interface between users and kernel
- ✓ It is a command interpreter and also has programming capability of its own.

File System:

Linux treats everything as a file including hardware devices. Arranged as a directory hierarchy.

- ✓ The top level directory is known as “root (/)”.

Linux Commands:

- **HELP**

\$ man [command name]

Purpose: provide all the details about command given. E.g. **\$ man ls**, **\$ man cat**

- **Date Time**

This command is used to display the current data and time.

Syntax: \$date

\$date +%H

Options: -

a =Abbrevated weekday.

A = Full weekday.

b =Abbrevated month.

B = Full month.

c = Current day and time.

C = Display the century as a decimal number.

d = Day of the month.

D = Day in „mm/dd/yy“ format

h =Abbrevated month day.

H = Display the hour.

l = Day of the year.

m = Month of the year.

M = Minute.

P = Display AM or PM

S = Seconds

T = HH:MM:SS format

u = Week of the year.

y = Display the year in 2 digit.

Y = Display the full year.

Z = Time zone.

Purpose: Prints the system date and time.

- **Calendar**

\$ cal

Purpose: Prints an ASCII calendar of the current month.

- **Current Working Directory**

\$ pwd

Purpose: Displays the absolute path to the current working directory.

- **Changing Directories**

\$ cd

Purpose: Changes directories.

i. **\$ cd**

Purpose: to your home directory (current directory).

ii. **\$ cd home /hitec/ cs**

Purpose: To an absolute path (from root directory).

iii. **\$ cd project /docs \$ cd Downloads \$ cd ./Downloads \$ cd**

Downloads/. \$ cd ./Downloads/. \$ cd Downloads/../Downloads

Purpose: To a relative path (from present working directory).

iv. **\$ cd ..**

\$ cd ~

Purpose: To a directory one level up.

v. **\$ cd -**

Purpose: To your previous working directory

- **cd /u*/f***

Moves you to the "/usr/fred" directory - if this is the only directory matching this wildcard pattern.

- **Listing Directory Contents**

\$ ls [options] [files_or_directories]

Purpose: Lists the contents of the current directory or a specified directory.

- i. **\$ ls**

Purpose: Without arguments lists the files and directories names in the current directory.

- ii. **\$ ls /**

Purpose: Lists the contents of the directory given as an argument.

- **ls -al**

This command provides a *long* listing of information about *all* files in the current directory. *This is probably the most used version of the ls command.*

- iii. **\$ ls -a /home /student**

Purpose: Includes so-called "hidden" files and directories whose names begin with a dot (.):

Note: Type **man ls** in shell to get Help about **ls** command (all available options of **ls** will be displayed).

- **Copying files and directories:**

\$ cp file destination

Purpose: Copy files and directories. More than one file may be copied at a time if the destination is a directory. (if source is a file and multiple destinations are there so these must be directories only one destination can be file)

Note:

If the destination exists and is a directory, the copy is placed there with the same name.

If the destination exists and is a file, the copy overwrites the destination file.

Examples

- **cp profile profile.bak**

This command copies your ".profile" to a file named ".profile.bak".

- **cp /usr/fred/Chapter1 /usr/mary**

This command copies the "Chapter1" file in "/usr/fred" to the directory named "/usr/mary". This example assumes that you have write permission in the "/usr/mary" directory.

- **Moving files and directories:**

\$ mv file destination

Purpose: Move files and directories from one place to another. More than one file may be moved at a time if the destination is a directory.

Examples mv Chapter1 Chapter1.bad

This command renames the file "Chapter1" to the new name "Chapter1.bad".

- mv Chapter1 garbage

This command **renames the file "Chapter1"** to the new name "garbage". (Notice that if "garbage" is a directory, "Chapter1" would be moved into that directory).

- mv Chapter1 tmp

This command moves the file "Chapter1" into the directory named "tmp".

- **Removing and creating files**

\$ rm filename

Purpose: Remove files

\$ mkdir dirname

Purpose: Create a new directory with the given name.

\$ rmdir -r dirname

Purpose: Removes specific directory

Examples

- mkdir tmp

This command creates a new directory named "tmp" in your current directory. (This example assumes that you have the proper permissions to create a new sub-directory in your current working directory.)

- mkdir memos letters e-mail

This command creates three new sub-directories (memos, letters, and e-mail) in the current directory.

- mkdir /usr/fred/tmp

This command creates a new directory named "tmp" in the directory "/usr/fred". "tmp" is now a sub-directory of "/usr/fred". (This example assumes that you have the proper permissions to create a new directory in /usr/fred.)

- `mkdir -p /home/joe/customer/acme`

This command creates a new directory named `/home/joe/customer/acme`, and creates any intermediate directories that are needed. If only `/home/joe` existed to begin with, then the directory "customer" is created, and the directory "acme" is created inside of customer.

\$ rmdir dirname

Purpose: Remove an empty directory.

Description The "rm" command is used to remove files and directories. (Warning - be very careful when removing files and directories!)

Examples

`rm Chapter1.bad`

This command deletes the file named "Chapter1.bad" assuming you have permission to delete this file).

- `rm Chapter1 Chapter2 Chapter3`

This command deletes the files named "Chapter1", "Chapter2", and "Chapter3".

- **`rm -i Chapter1 Chapter2 Chapter3`**

This command prompts you before deleting any of the three files specified. The -i option stands for *inquire*.

You must answer y (for yes) for each file you really want to delete. This can be a safer way to delete files.

- `rm *.html`

This command deletes all files in the current directory whose filename ends with the characters ".html".

- `rm index*`

This command deletes all files in the current directory whose filename begins with the characters "index".

- `rm -r new-novel`

This command deletes the directory named "new-novel". This directory, and all of its' contents, are erased from the disk, including any sub-directories and files.

Viewing and editing a Text File

- Using touch to **create** a blank text **file**: `$ touch NewFile.txt`.
- Using cat to **create** a new **file**: `$ cat NewFile.txt. ...` (write anything in your file CNTR+D to save and CTRL+Z to exit)
- Simply using > to **create** a blank text **file**: `$ > NewFile.txt`.

\$ cat [option] [filename]

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i. **\$ cat < filename**

Purpose: It will display the contents of the file *filename*.

ii. **\$ cat > file1**

Purpose: “>” is called output redirection operator. It permits user to write in *file1*. If *file1* already exists then it overwrites the contents of the *file1*.

E.g. write any word/phrase after giving command **\$ cat > file1** like “Success is not a destination.”

Press **[Ctrl+d]**

Note: The above command creates the file called *file1* if *file1* does not exist and allows user to write text there. Press Ctrl+d (To save and exit).

Tasks

Show **Simulation** results after performing the following tasks by using suitable commands in Linux

1. Verify that you are in your home directory.
2. Make the directory *adir* using the command: `mkdir adir` if problem occurs, use `sudo mkdir adir` and then give password
3. List the files in the current directory to verify that the directory *adir* has been made correctly.
4. Change directories to *adir*.
5. Verify that you have succeeded in moving to the *adir* directory.
6. Make file *testfile*.
7. Verify that the file *testfile* exists.
8. List the contents of the file *testfile* to the screen.
9. Make a copy of the file *testfile* under the name *secondfile*.
10. Verify that the files *testfile* and *secondfile* both exist.
11. List the contents of both *testfile* and *secondfile* to the monitor screen.
12. Delete the file *testfile*.
13. Verify that *testfile* has been deleted.
14. Clear the window.
15. Rename *secondfile* to *thefile*.
16. Copy *thefile* to your home directory.
17. Remove *thefile* from the current directory.

18. Verify that *thefile* has been removed.
19. Copy *thefile* from your home directory to the current directory.
20. Verify that *thefile* has been copied from your home directory to the current directory.
21. Change directories to your home directory.
22. Verify that you are in your home directory.
23. Verify that a copy of *thefile* is in your home directory.
24. Remove *thefile* from your home directory.
25. Remove *thefile* from the directory *adir*.
26. Remove the directory *adir* from your home directory with the command. `rmdir adir`
27. Verify that *thefile* and *adir* are gone from your home directory.

Conclusion:

Lab Instructor

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